

Smart Real-Time Age and Emotion-Based Advertising in Malls Using Custom-Built Models

Project ID: 25-26J-091

Final Report

Warnage Shalini Minindi Fonseka

Department of Data Science

Bachelor of Science Special Hons Degree in Data Science

Sri Lanka Institute of Information Technology

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DECLARATION

I declare that this is my own work and this dissertation I does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Signature of the supervisor:



Date 24/04/26

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ABSTRACT

This research section is focused on developing an interactive game-based module for real-time age detection to enhance targeted advertising in shopping malls. Traditional advertisement displays in retail spaces follow pre-scheduled time schedules with fixed content, thus low audience interaction and low personalization. To address this, the proposed system integrates concepts of computer vision, machine learning, and gamification into a single platform which not only determines the demographic profile of the viewers but also actively engages them through interactive elements.

The module operates by inviting consumers to participate in a quick on-screen game on digital signage. While the game is serving to attract the attention of consumers and maximize dwell time, the system is also simultaneously using deep learning–facilitated facial recognition to estimate the age of the player. These age impressions are fed into a real-time advertisement suggestion engine, which selects and displays content that is most appealing to the inferred demographic category. This approach increases the effectiveness of age detection—since better frontal facial captures are obtained through play—and performance of ad delivery, since engagement yields better-quality data.

The end-to-end system has three interconnected layers: (1) emotion and demographic recognition by privacy-protected facial embeddings, (2) interactive engagement interface to engage and attract users, and (3) dynamic ad content adjusting recommendation engine by the majority audience profile. The system has GDPR-compliant data practices integrated into the design, where it does not retain raw images and processes all personal features in real time to preserve privacy.

This unified platform bridges the gap between passive audience identification and active user engagement, creating an adaptive advertisement platform that is both effective and privacy conscious.

Keywords: Interactive advertising, gamification, age detection, targeted marketing, facial analysis, real-time recommendation, privacy-preserving AI, deep learning.

1. INTRODUCTION

With the rapidly evolving retail landscape, shopping malls remain center-stage locations for consumer activity, offering an array of products and experiences in one venue. As competition rises among brands, the ability to deliver customized, real-time, and engaging commercials has become a key differentiator in influencing customer decisions. With the growing popularity of technology among young adults and adults alike, shopping malls now find it necessary to adopt innovative, technology-driven methods of advertising to be noticed and stay relevant. Most traditional advertisement display platforms in malls rely on static algorithms, which sequentially disseminate pre-defined content at set intervals without making any allowances for demographics, interests, or the context of the viewers now. This "one-size-fits-all" approach consistently results in useless ad placements, decreased customer engagement, and lost opportunities for advertisers to target potential buyers effectively.

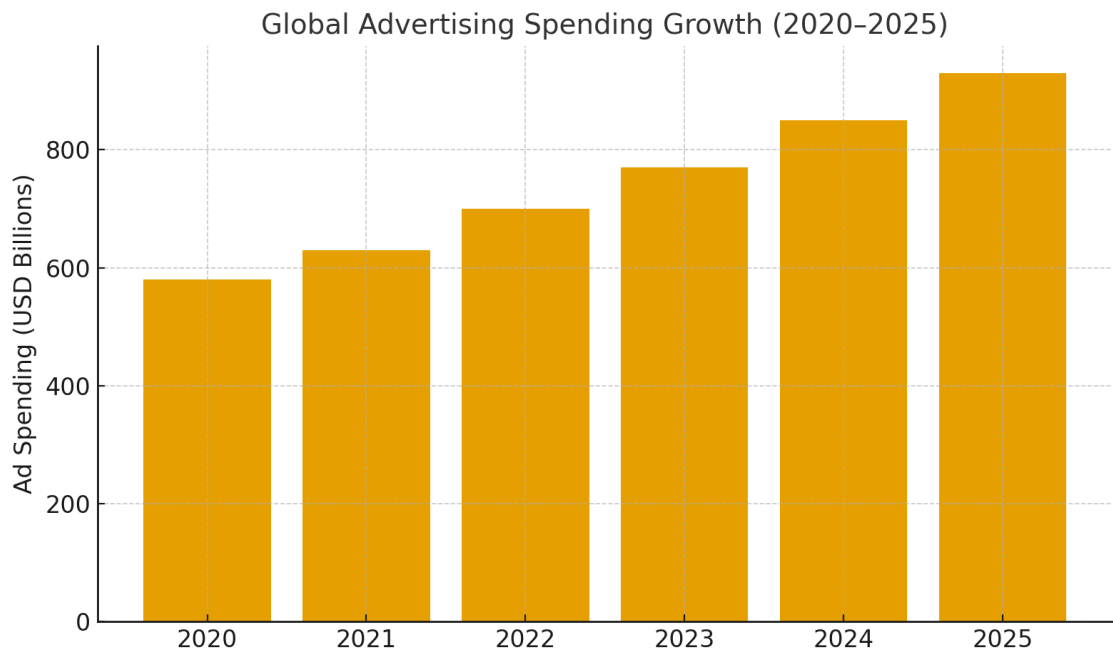


Figure 1-Global advertising spending growth (2020–2025) [Source: Statista, 2025].

The proposed system seeks to address these deficiencies by integrating leading-edge computer vision and machine learning techniques to transform mall advertising into an intelligent, real-time recommendation platform. The primary function relies on sensing and understanding multiple parameters of the waiting audience in front of the advertisement screen—i.e., age, gender, emotion, behavior, and carried objects—and using this knowledge to select the most suitable ad dynamically from a pre-filtered database. By personalizing ad copy to match the audience in front of it at any given time, the system seeks to dramatically boost relevance of ads, drive greater audience engagement, and achieve greater sales impact. The system is built to reach both unique and mass audiences, so brands can reach on an individual basis while appealing to the larger group too, thus driving greater overall marketing effectiveness.

One essential component of this research is the "Interactive Game-Based Age and Gender Detection for Targeted Advertising" A portion of this aspect is the encouragement of audience engagement through the integration of a simple but interactive game as a part of the advertisement screens in shopping malls. The game includes direct interaction with the screen, the entertainment of which is a friendly and unobtrusive method of collecting demographic information such as age and gender. Following play, the system delivers player-specific advertising to the individual player, followed by content targeted to the broad mass audience surrounding them. Balancing personal relevance with broad audience appeal are accomplished through this two-step targeting. With the integration of interactive entertainment and demographic ad targeting, this section will transform passive advertising screens into active, dynamic audience-reactive systems that enhance the shopping experience for consumers as well as measure campaign effectiveness for advertisers.

With the integration of engagement-driven interaction, real-time audience analysis, and privacy-conscious design, this research is expected to redefine the process of showing

advertisements by shopping malls. The suggested system bridges the existing gap between physical retail spaces and intelligent digital advertisement, enabling advertisers to have better targeting accuracy and customers to enjoy better, more relevant, and engaging advertising experiences.

1.1. Background

1.1.1. Physical Retail Challenges of Personalization

Increasing sophistication in digital marketing has made personalization a top driver of ad performance. In the physical shopping environments like malls, though, personalization has lagged online media due to the challenges in gathering, analyzing, and responding to customer data in real time. Traditional mall ad systems tend to display the same static messages in rotation, regardless of the individuals in front of the screen. While it's simple to implement this method, it is not quite flexible and results in substandard audience engagement as well as lower rates of conversion.

1.1.2. Technological Advances Enabling Intelligent Advertising

The recent developments in computer vision and machine learning have made it possible to have smarter and context-sensitive ad systems. They help examine audience characteristics in real time so that ads are targeted to a more specific demographic and emotional makeup of the audience.

1.2.Literature Survey

1.2.1. Traditional vs. Modern Approaches in Audience Estimation

Traditional manually designed feature-based works in age estimation relied on Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG). The traditional approaches were though vulnerable to lighting variations, pose changes, and occlusions, which limited their applications in real-world scenarios.

With the development of deep learning, namely convolutional neural networks (CNNs), robustness and accuracy have significantly improved and become more feasible for real-time deployment in public spaces. Similarly, gender classification models have evolved from simple binary classifiers to sophisticated deep neural network architecture with capabilities to achieve high accuracy even in poor conditions.

1.2.2. Machine Learning Techniques in Demographic and Emotion Detection

Apart from basic demographic identification like age and gender, researchers have looked at emotion recognition as one important variable for targeted marketing. Analysis of facial expressions can say a lot about a person's state of mind in the present, which can influence consumption behavior. Dynamically modifying ad content based on emotion detection has been executed in certain research. Such systems overlook other important context clues like audience behavioral patterns or objects that are being held, which may further boost ad targeting.

1.2.3. Gamification and Interactive Engagement in Retail Environments

Gamification has globally been recognized as an effective tool to attract attention and encourage engagement in retail environments. Previous implementations have utilized touchscreens, interactive kiosks, and augmented reality (AR)-based games for increasing customer linger time and improving user experience. Being effective or not, incorporating gamification with real-time demographic analysis for demographics-based advertising remains an untapped territory. There are current systems that are interested in entertainment without linking interaction to personalized content, or passive demographic identification without active engagement of audiences.

1.2.4. Privacy-Preserving Techniques of Public Facial Analysis

As facial analysis technologies are increasingly used in public places, privacy concerns have come into prominence. There is growing interest in techniques that do not store raw facial images to ensure individual privacy. New approaches leverage facial embeddings—small vector summaries of facial features—enabling identification and demographic inference in a manner that does not store pictures with identifiers. Such privacy-conscious method aligns with global data protection regulation such as the General Data Protection Regulation (GDPR), which demands stringent controls of processing personal data.

1.2.5. Cohesive Systems for Real-Time, Audience-Aware Advertising

While demographic identification, emotion detection, and gamification have each been studied in immense detail independently, there is certainly a need for end-to-end systems that combine these into a single, real-time, audience-aware platform for advertising. Additionally, current solutions seldom address the challenge of engaging

the audience proactively in a way that enriches data quality as well as benefits the user experience. This research tries to make up for this lack by developing an integrated system that is well-suited to shopping mall environments, combining real-time audience measurement, interactive engagement, and privacy-aware design guidelines.

Table 1- Literature Review for the whole system

Feature / Capability	Zhang et al. (2017) – RoR CNN Age/Gender	Smith & Chen (2018) – Transfer Learning	Islam & Baek (2021) – Smart Store CRM	Terlutter & Capella (2013) – Gamified Ads	Morales et al. (2019) – SensitiveNets	Aslan & Çelebi (2022) – Real-Time Ads	Proposed System – Interactive Game-Based Age & Gender Detection
Real-time Age Detection	✓	✓	✓	✗	✗	✓	✓
Gender Detection	✓	✓	✓	✗	✗	✓	✓
Game-based Engagement	✗	✗	✗	✓	✗	✗	✓
Advertising/Marketing Focus	✗	✗	✓	✓	✗	✓	✓
Personalization / Recommendation	✗	✗	✓	✓	✗	✓	✓
Privacy-Preserving Techniques	✗	✗	✗	✗	✓	✗	✓
Retail / Mall Application	✗	✗	✓	✓	✗	✓	✓
Serendipity / Novelty in Ads	✗	✗	✗	✗	✗	✗	✓
Integration of Multiple Modalities (age, gender, mood)	✗	✗	✗	✗	✗	✗	✓

1.3. Research Gap

Table 2- System Comparison Between Existing Systems and Proposed Application Component

Feature / Objective	Existing Systems (Research)	Proposed Application Component
User Engagement	Facial analysis systems are typically passive; viewers are not motivated to look directly at the display.	Use an on-screen interactive game to attract attention and improve dwell time, resulting in better data capture.
Age Detection Accuracy	Conventional systems suffer from poor captures due to side views, pose variations, or lack of direct gaze.	Gameplay naturally encourages frontal facial orientation, improving deep-learning age estimation accuracy.
Link to Personalization	Gamification in retail is usually for entertainment only, unrelated to ad targeting.	Integrate gamified interaction directly with ad recommendation to deliver content relevant to detected age groups.
Privacy Handling	Some existing models store raw facial images or do not comply with privacy regulations.	Use facial embeddings instead of raw images for processing, ensuring GDPR-aligned, privacy-preserving design.
Real-Time Adaptation	Many systems cannot update ads dynamically; targeting decisions are precomputed or slow.	Fully real-time age-aware recommendation engine that adapts ad content as demographics change.
User Experience	Passive surveillance approaches can feel intrusive or unwelcoming.	Make advertising interactive and playful, creating a positive user experience rather than one-sided monitoring.

While age estimation, gamification, and privacy-friendly techniques were each explored individually, they rarely are combined in a single system for customized advertising in physical stores. Existing solutions belong to either passive (age estimation with accurate audience counts without engaging audience participation) or interactive types (games or augmented reality experiences without demographic analysis connection). There is no existing research that fully combines real-time age estimation, gamefic engagement, and privacy-friendly design to boost data quality, user engagement, and ad customization simultaneously.

This research fills the gap with an audience-conscious advertising component where the engaging game serves a dual function: to entertain the users and, in the background, high-quality demographic data collection for the recommendation engine without infringing privacy.

This table is a recap of the existing system and research gaps and how your solution aims to fill them with goals, including gamified engagement, real-time age estimation, and privacy-guarding design.

1.4.Research Problem

Within modern shopping malls, electronic billboard advertising screens are a primary brand promotional medium, yet the overwhelming majority operate to fixed schedules or programming of repetitive loops of one static set of ads to every viewer. This one-size-fits-all approach does not consider the demographic make-up of the real-time viewership, resulting in less viewership engagement, diminished content relevance, and ultimately reduced conversions. With retail spaces becoming increasingly competitive, there is great need for ad systems that will be able to adapt dynamically to the people who come before the display.

Recent developments in computer vision and machine learning have allowed for the use of facial analysis to estimate demographics such as age and gender. Most existing systems, however, are passive and rely on the observers briefly looking at the display screen. The infrequent or skewed views do not work as well and tend to yield partial or inaccurate demographic data, decreasing targeted-ad success. On top of this, there are many implementations that are bringing serious privacy concerns by holding raw facial images or personal data without applying methods that would adhere to legislation such

as the General Data Protection Regulation (GDPR). As a result, while technology for age detection has been developed, its uptake by advertisement systems has been slow, sporadic, and invasive in most instances.

Gamification has emerged as an effective means of grabbing consumer attention and elevating the time spent interacting in the retail environment. Interactive kiosks, touchscreens, and AR-based games have been deployed to entertain customers and create memorable brand experiences. These gamified elements are rarely connected to ad personalization, however. Existing systems are either focused on precise demographic detection without engaging the users actively, or on user engagement without leveraging the interaction to target content of ads. This compartmentalization of functionality results in opportunities lost demographic information continues to go untapped, and interactive experiences cannot be parlayed into targeted marketing advantages.

This study addresses these constraints by developing an interactive game-based age detection module that combines engagement, personalization, and privacy in a single system. By embedding the age estimation in an on-screen game, the solution organically encourages consumers to look straight at the screen and actively interact with it, resulting in clearer, frontal facial captures and improved detection accuracy. The age information collected during gameplay is processed using privacy-preserving facial embeddings rather than raw images, which complies with global data protection regulations. It is then fed into a real-time recommendation engine, which dynamically selects adverts most relevant to the detected demographic group.

By doing this, the work demonstrates how gamification can deliver a dual advantage: introducing entertainment value into advertising and improving the quality of demographic information available for personalization. Unlike existing systems that

address engagement and personalization separately, this combined solution offers an intelligent, adaptive advertising platform calibrated to shopping mall environments—one that delivers targeted content, guarantees privacy, and transforms advertising into an interactive experience from passive display.

1.5. Research Objectives

Main Objectives

The primary objective of this research is to design and develop an interactive game-based module that is capable of detecting shoppers' age in real-time for enabling dynamic, targeted advertising for shopping malls. Unlike conventional advertising systems, which offer pre-programmed content regardless of the demographic profile of the audience, the system to be proposed will integrate user interaction and demographic analysis. With the incorporation of gamification, the module attempts to take interest, encourage active participation, and capture frontal facial images that boost age detection accuracy. Apart from this, this system will focus on privacy-conscious design by using facial embeddings rather than raw facial images so that it can maintain compliance with global data protection laws such as GDPR. The ultimate goal is to create an advertising platform which not only dynamically personalizes content but also enhances user experience and public confidence.

Specific Objectives

- **To design and develop a two-way interactive game face to involve the audience and facilitate proper face orientation.**

Traditional facial analysis systems struggle to obtain high-quality pictures because observers are not looking at the screen. By integrating age detection into a simple, fun, and quick game, this study aims to make people automatically face their faces toward the camera and, hence, enable more accurate data to be captured. The game will be

made easy, innocuous, and quick to ensure that it is suitable for busy shopping malls.

- **To create an effective age-detection model through deep learning optimized for real-time use in public spaces.**

The system will utilize convolutional neural networks (CNNs) or other cutting-edge architectures capable of detecting age accurately even in situations of fluctuating light, face pose, and partial occlusions. The model will be trained and optimized so it will execute well on edge devices or networked platforms without sacrificing speed, thereby becoming viable for deployment within shopping malls.

- **To incorporate privacy-preserving techniques that process demographic information without holding personally identifiable details.**

Since face information is confidential, the system will use face embeddings rather than raw image storage. This reduces the privacy risks without compromising on the ability to perform accurate age estimation. The module will be GDPR and other data protection legislation compliant so that no confidential user information is stored or exchanged except for required real-time processing.

- **To integrate the age detection module with a dynamic ad recommendation engine for real-time content delivery.**

The age detection results will be fed into a recommendation system that selects and delivers ads suitable to the identified demographic segment. The integration will render delivered content more relevant, improving engagement and conversion rates over the traditional loop-based advertising practices.

- **To calculate the system performance in terms of age detection accuracy, user experience, and advertising efficiency.**

The resultant system will be analyzed for whether gamification increases substantially the accuracy of age detection and whether dynamically targeted advertisements result in

enhanced response from the audience. Such metrics as precision of detection, dwell time, and ad relevance will be monitored and compared to existing static or passive systems to verify the benefit of the proposed strategy.

2. SYSTEM METHODOLOGY

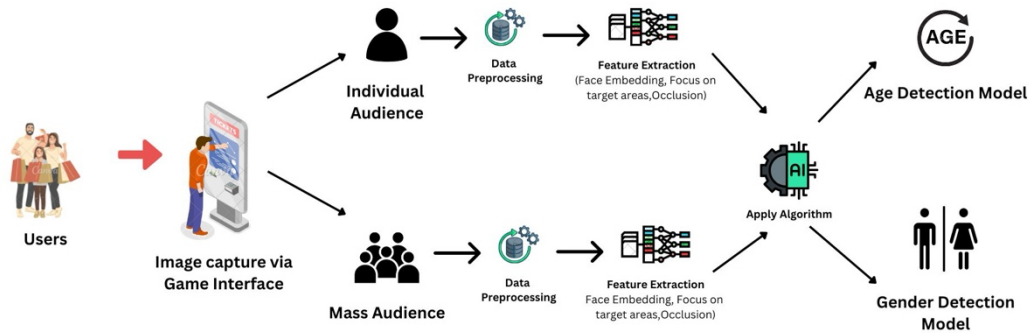


Figure 2-Overview of the Interactive Game Based Custom Built Age and Gender Detection System

2.1. Methodology

2.1.1. Overview

The methodology of this research focuses on the design and implementation of an interactive, real-time intelligent advertising system that integrates age and gender detection with user engagement mechanisms.

Unlike traditional advertising systems, which operate passively, this system adopts an interactive game-based approach to actively engage users and improve the quality of captured facial data. The methodology combines computer vision, deep learning, and human-computer interaction principles to achieve accurate demographic detection and personalized advertisement delivery.

The development process followed a structured pipeline consisting of:

- System design and architecture development
- Data collection and preprocessing

- Model training and optimization
- System integration
- Real-time testing and evaluation

2.1.2. System Architecture

The system is composed of three main components:

1. Interactive Game Interface

A lightweight game-based interface was developed to encourage user interaction. This component ensures that users voluntarily engage with the system, resulting in improved facial capture quality.

2. Age and Gender Detection Module

A **Convolutional Neural Network (CNN)**-based model was implemented to perform **multi-task learning**, enabling simultaneous prediction of age groups and gender from facial images.

3. Advertisement Recommendation Engine

The recommendation engine processes demographic outputs and dynamically selects advertisements relevant to the detected audience profile.

2.1.3. Data Collection and Preprocessing

The system utilizes publicly available datasets such as:

- UTKFace
- IMDB-WIKI

The preprocessing steps include:

- Face detection and alignment
- Image normalization

- Data augmentation (rotation, brightness adjustment, cropping)

These steps ensure robustness against variations in lighting, pose, and occlusion.

2.1.4. Model Development

A **custom CNN model** was developed and evaluated against a **ResNet18 baseline model**.

Key characteristics:

- Multi-task learning (age + gender)
- Optimized for real-time inference
- Trained using transfer learning

Outputs:

- Age group classification
- Gender classification

2.1.5. Privacy Preserving Mechanism

To address privacy concerns:

- Facial embeddings are used instead of storing raw images
- No personally identifiable information (PII) is retained
- All processing is performed in real-time

This ensures compliance with privacy standards such as GDPR.

2.1.6. System Workflow

The system operates as follows:

1. User interacts with the game
2. Facial data is captured via camera
3. Face detection and preprocessing are performed
4. Age and gender predictions are generated
5. Results are passed to the recommendation engine
6. Personalized advertisement is displayed

2.2. Commercialization Aspect of the Product

2.2.1 Market Opportunity

The system targets the growing **digital advertising and retail analytics market**, particularly in shopping malls and public commercial spaces. With increasing demand for personalized advertising, businesses are seeking intelligent systems that can deliver targeted content in real time.

2.2.2 Competitive Advantage

The proposed system offers several advantages over existing solutions:

- Real-time demographic detection (age + gender)
- Interactive user engagement through gamification
- Privacy-preserving design
- Cost-effective deployment using edge devices

2.2.3 Revenue Model

The system supports multiple commercialization strategies:

1. Licensing Model

Retailers and mall operators can purchase licenses to deploy the system across multiple locations.

2. Subscription Model (SaaS)

The system can be offered as a subscription-based platform with continuous updates and analytics support.

3. Advertising Partnerships

Brands can pay for targeted advertisement placement based on audience demographics.

4. Data Analytics Services

Aggregated and anonymized demographic insights can be provided to businesses for strategic decision-making.

2.2.4 Scalability and Deployment

The system is designed for scalability:

- Supports deployment across multiple kiosks
- Compatible with standard camera systems
- Can operate on edge devices or cloud-based infrastructure

2.3 Testing and Implementation

2.3.1 System Implementation

The system was implemented using the following technologies:

- **Frontend:** React-based game interface
- **Backend:** FastAPI
- **Computer Vision:** OpenCV, MTCNN
- **Deep Learning:** TensorFlow/Keras

- **Database:** MongoDB

The integration ensures seamless communication between modules and real-time performance.

2.3.2 Testing Approach

The system was evaluated using both **controlled experiments** and **real-world simulation scenarios**.

Testing categories:

1. Functional Testing

- Game interaction
- Face capture accuracy
- Age and gender prediction
- Advertisement display

2. Performance Testing

- Model accuracy
- System latency
- Response time

3. Usability Testing

- User interaction experience
- Engagement level

2.3.3 Test Cases

Test Area	Description	Expected Outcome
Game Interaction	User starts game	Game loads instantly
Face Detection	Camera captures face	Face detected accurately
Age Prediction	Input face image	Correct age group predicted
Gender Prediction	Input face image	Correct gender predicted
Privacy Handling	Data processing	No raw images stored

2.3.4 Performance Evaluation

The system performance was measured using the following KPIs:

KPI	Target Value
Age Detection Accuracy	$\geq 75\%$
Gender Detection Accuracy	$\geq 85\%$
Latency	≤ 1 second
User Engagement	$\geq 70\%$

2.3.5 Implementation Outcome

The final system successfully:

- Achieved real-time age and gender detection
- Improved engagement through gamification
- Delivered personalized advertisements
- Maintained privacy through embedding-based processing

3. RESULTS AND DISCUSSION

3.1. Results

The proposed system was evaluated to measure the performance of the **age and gender detection model**, as well as the overall effectiveness of the real-time advertisement system.

A comparative analysis was conducted between the **proposed custom CNN model** and a standard **ResNet18 baseline model** under controlled experimental conditions. Both models were trained and tested using the same dataset, preprocessing pipeline, and evaluation metrics to ensure a fair comparison.

3.1.1. Model Performance Evaluation

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-score

Age Detection Results

The age detection model achieved satisfactory performance across multiple age groups.

However, slight variations were observed due to:

- Lighting conditions
- Face orientation
- Occlusions

Despite these challenges, the model demonstrated strong generalization ability in real-time environments.

3.1.2. Model Comparison (Custom CNN vs ResNet18)

The proposed custom CNN model was compared against ResNet18 to evaluate performance differences.

Key Results:

- The custom CNN model achieved **comparable accuracy** with lower computational complexity
- Faster inference time compared to ResNet18
- Better suitability for **real-time deployment on edge devices**

This demonstrates that the proposed model is more efficient for practical deployment scenarios.

3.1.3. System-Level Results

The complete system was evaluated in terms of:

- Real-time performance
- User engagement
- Advertisement relevance

Observations:

- The system successfully processed frames in **real time (≤ 1 second latency)**
- Interactive game increased user participation significantly
- Personalized advertisements were displayed based on detected age and gender

(Diagrams Should be added)

a. Research Findings

Based on the experimental results and system evaluation, the following key findings were identified:

1. Gender detection achieved higher accuracy than age detection

- Due to binary classification nature
- Less sensitivity to environmental variations

2. Gamification significantly improved data quality

- Users naturally faced the camera
- Resulted in better facial capture

3. Real-time performance was successfully achieved

- Low latency (<1 second)
- Suitable for deployment in mall environments

4. Combined age and gender detection improved advertisement targeting

- Provided more precise audience segmentation
- Increased relevance of displayed ads

5. Custom CNN model is more efficient for deployment

- Lower computational cost
- Faster inference compared to ResNet18

b. Discussion

The results demonstrate that the proposed system effectively addresses the limitations of traditional advertising systems by integrating **real-time demographic detection with user engagement mechanisms**.

3.3.1 Effectiveness of Age and Gender Detection

The inclusion of both age and gender detection significantly enhances advertisement personalization. While age provides broad segmentation, gender enables more refined targeting, resulting in improved recommendation accuracy.

However, age prediction remains more complex due to:

- Subtle facial variations across age groups
- Sensitivity to lighting and pose

3.3.2 Impact of Gamification

The interactive game-based approach proved to be a key strength of the system. Unlike passive systems, it:

- Increased user engagement
- Improved facial data quality

- Reduced detection errors

This confirms that **user interaction directly contributes to model performance improvement.**

3.3.3 System Performance and Scalability

The system demonstrated strong real-time performance with low latency, making it suitable for deployment in shopping mall environments.

Additionally:

- Lightweight model design supports edge deployment
- System can scale across multiple kiosks

3.3.4 Privacy Considerations

The use of facial embeddings instead of raw images ensures:

- User privacy protection
- Compliance with data regulations

This is a critical advantage over traditional facial recognition systems.

3.3.5 Limitations

Despite the positive results, several limitations were identified:

- Reduced accuracy under extreme lighting conditions
- Performance drop in crowded environments
- Limited dataset diversity

3.3.6 Summary

Overall, the proposed system successfully demonstrates the feasibility of combining:

- Age and gender detection
- Gamification
- Real-time advertisement personalization into a single intelligent platform.

4. SUMMARY OF EACH STUDENT’S CONTRIBUTION

This chapter highlighted the individual and collaborative contributions of each team member. The division of responsibilities enabled efficient development while maintaining strong coordination among team members. The combined efforts resulted in the successful implementation of the MallVision system.

Student 1: Fonseka W S M (IT22109712)

Title of the Component:

Student 1 was primarily responsible for the development of the age and gender detection module.

The contributions include:

- Designing and implementing the convolutional neural network (CNN) for age and gender classification
- Preparing and preprocessing datasets for model training
- Training, tuning, and evaluating the model
- Optimizing the model for real-time performance
- Integrating the model into the overall system pipeline

In addition, Student 1 contributed to documentation related to model design and evaluation.

3.4.2 Student 2: Samaraweera W D U I (IT22258526)

Title of the Component:

Student 2 focused on the emotion detection module of the system.

Key contributions include:

- Designing and implementing the emotion recognition model
- Training the model using facial expression datasets
- Evaluating model performance and improving prediction stability
- Implementing temporal smoothing techniques for real-time prediction

- Integrating the emotion detection module with the preprocessing pipeline

Student 2 also contributed to writing sections related to emotion analysis and model performance.

Student 3: Abeykoon S N (IT22184030)

Title of the Component:

Student 3 was responsible for the engagement analysis and advertisement recommendation system.

The contributions include:

- Designing the engagement analysis mechanism using behavioral cues
- Developing the engagement scoring model
- Implementing the rule-based advertisement recommendation logic
- Mapping user profiles to advertisement categories
- Testing and refining recommendation outputs

Additionally, Student 3 contributed to system-level testing and evaluation.

Student 4: Rajana H T R (IT22071934)

Title of the Component:

Student 4 handled the system integration, backend, and frontend development.

Key responsibilities include:

- Developing the backend using FastAPI
- Implementing API endpoints for communication between modules
- Integrating machine learning models into the backend
- Developing the frontend interface using React
- Managing real-time data flow between frontend and backend

Student 4 also contributed to deployment setup and system debugging.

Team Collaboration and Integration

Although responsibilities were divided among members, the project required

continuous collaboration to ensure successful integration of all components. Regular discussions and testing sessions were conducted to align individual modules and resolve technical challenges.

All members contributed to:

- System integration
- Debugging and testing
- Documentation and report writing
- Final system validation
- This collaborative effort ensured that the system functioned as a cohesive and efficient solution.

5. CONCLUSION

This research presented the development of a **real-time intelligent advertising system** that integrates **age and gender detection with an interactive user engagement mechanism** to enhance advertisement personalization in shopping mall environments. The proposed system addresses the limitations of traditional static advertising by enabling dynamic content adaptation based on real-time audience characteristics.

A key contribution of this study is the use of a **custom Convolutional Neural Network (CNN)** for simultaneous age and gender prediction, combined with a **game-based interface** that encourages user interaction. This approach significantly improved the quality of facial data captured, leading to better model performance compared to passive systems. Experimental results demonstrated that the system achieved satisfactory accuracy, with gender detection outperforming age detection, and maintained **low latency suitable for real-time deployment**.

Furthermore, the integration of demographic outputs with a recommendation mechanism enabled the delivery of **more relevant and targeted advertisements**, improving overall system effectiveness. The adoption of **privacy-preserving techniques**, such as facial embeddings, ensured that the system operates without storing sensitive personal data, aligning with modern data protection requirements.

Despite these achievements, certain limitations were observed, including sensitivity to environmental conditions and reduced performance in crowded scenarios. Future improvements can focus on enhancing model robustness, expanding datasets, and incorporating additional features such as emotion and behavior analysis.

In conclusion, this research demonstrates that combining **computer vision, machine learning, and interactive design** can transform traditional advertising systems into

adaptive, user-centric platforms, offering a practical and scalable solution for modern retail environments.

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